



National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Department of Computing

CS220: Database Systems

Class: BSAI – 2A

Lab 02: SQL Operators

CLO-2: Formulate SQL queries to retrieve information from a relational database.

CLO 3: Build SQL (DDL &DML) queries to retrieve information from a relational database.

Date: 28 January 2026

Time: 10:00 - 12:50

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Introduction

Structured Query Language (SQL) was developed at IBM San Jose Research Laboratory as a part of System R project. It is a declarative query language for querying a relational database. It also includes features for defining the structure of the data, for inserting and modifying data in the database, and for specifying security constraints. It is relational complete (it supports all six core relational algebra operations). SQL commands can be classified into three groups DDL, DML & DCL.

Objectives

After performing this lab students should be able to:

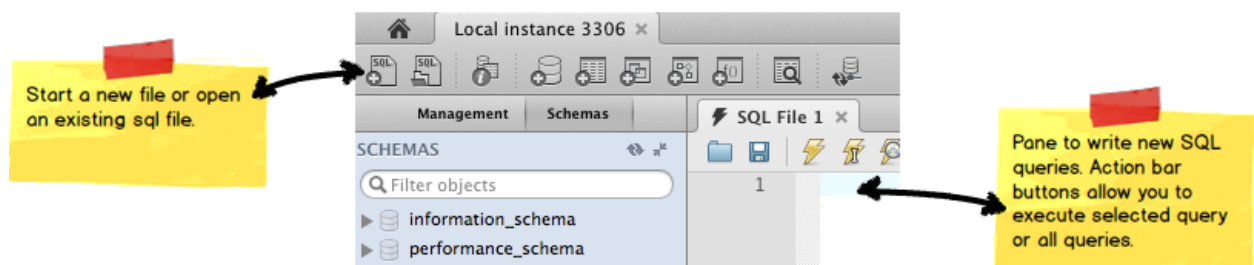
1. Design SQL queries to retrieve data using SELECT clause and various associated operators.
2. Design SQL queries with FROM & WHERE clause.
3. Execute SQL queries over MySQL using MySQL Workbench.

Tools/Software Requirement

- MySQL Community Server
- MySQL Workbench
- Sakila Database

Description

1. Open MySQL Workbench and open the default connection instance.
2. A new query window would open from where you can write and execute queries.



3. You can save the query file and can also add comments using # symbol.
4. On executing queries, results are displayed in the lower part of the screen.
5. Error or success messages are displayed in action output pane at the bottom.
6. Try running few SQL queries modeled during the lectures to get it going.
7. Continue playing with the Workbench and SQL queries till you are comfortable with the querying mechanism and have learnt the shortcuts to execute queries.



1. SQL Basic Query Structure

The SELECT clause

The most common use of the SQL commands is the selection of data from the tables located in a database. This can be achieved through SELECT command. We need to SELECT information FROM a table. Hence we have the most basic SQL query structure comprising of:

- SELECT
- FROM
- WHERE

The syntax for SELECT clause is as follows:

```
SELECT "column_name(s)" FROM "table_name(s)";
```

Always specify the name of the database in which a table/relation is present through which data is to be retrieved.

*e.g. Select column_name
 From Sakila.table_name*

There are three ways we can retrieve data from a table:

- Retrieve one column
- Retrieve multiple columns
- Retrieve all columns (Use *)

The select clause can contain arithmetic expressions involving the operation, +, -, *, and /, and operating on constants or attributes of records (tuples).

The FROM clause

The FROM clause can allow to select attributes from single table or multiple tables. When multiple tables are applied, it combines the records from the two or more tables listed and presents every possible combination of the listed attributes in SELECT clause.

This is not very useful until some filtering condition is applied. This is achieved through using the WHERE clause.

The WHERE clause

We can use the WHERE clause to filter the result set based on certain conditions. The syntax for using WHERE in the SELECT statement is as follows:

```
SELECT "column_name(s)"  
FROM "table_name(s)"  
WHERE "condition";
```

"Condition" can include a single comparison clause (called simple condition) or multiple comparison clauses combined together using AND or OR operators (compound condition). Conditions can include other operators like IN, BETWEEN, DISTINCT etc shown in Table 1:

Operator	Description	Example
=	Equal to	Author = 'Alcott'
<>	Not equal to (most DBMS also accept != instead of <>)	Dept <> 'Sales'
>	Greater than	Hire_Date > '2012-01-31'
<	Less than	Bonus < 50000.00
>=	Greater than or equal	Dependants >= 2



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<=	Less than or equal	Rate <= 0.05
BETWEEN	Between an inclusive range	Cost BETWEEN 100.00 AND 500.00
LIKE	Match a character pattern	First_Name LIKE 'Will%'

Table 1: List of Basic Operators available in SQL

2. Ordering data

The order of rows returned in a query result is undefined. The ORDER BY clause can be used to sort the rows. This clause comes last in the SELECT statement. ASC at the end of the ORDER BY clause specifies ascending order where as DESC specifies descending order. ASC is the default order. The syntax for an ORDER BY statement is as follows:

```
SELECT "column_name"  
FROM "table_name"  
[WHERE "condition"]  
ORDER BY "column_name" [ASC, DESC];
```

3. The Wildcard operators

There are times when we want to match on a string pattern. To do that, we will need to employ the concept of wildcard. In SQL, there are two wildcards:

- % (percent sign) represents zero, one, or more characters.
- _ (underscore) represents exactly one character.

Wildcards are used with the LIKE keyword in SQL.

Below are some wildcard examples:

- 'A_Z': All string that starts with 'A', another character, and end with 'Z'. For example, 'ABZ' and 'AZZ' would both satisfy the condition, while 'AKKZ' would not (because there are two characters between A and Z instead of one).
- 'ABC%': All strings that start with 'ABC'. For example, 'ABCD' and 'ABCABC' would both satisfy the condition.
- '%XYZ': All strings that end with 'XYZ'. For example, 'WXYZ' and 'ZZXYZ' would both satisfy the condition.
- '%AN%': All strings that contain the pattern 'AN' anywhere. For example, 'LOS ANGELES' and 'SAN FRANCISCO' would both satisfy the condition.
- '_AN%': All strings that contain a character, then 'AN', followed by anything else. For example, 'SAN FRANCISCO' would satisfy the condition, while 'LOS ANGELES' would not satisfy the condition.

Lab Task

Using Sakila Database

Formulate SQL queries for the following needs and execute them on Sakila database.



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1. Write a query to Select column “actor_id” from the table “actor” which already exists in the “Sakila” database where the “actor_id=58”. (Observe the output when you execute the queries.)

QUERY:

```
1 • select actor_id from actor where actor_id=58
2
```

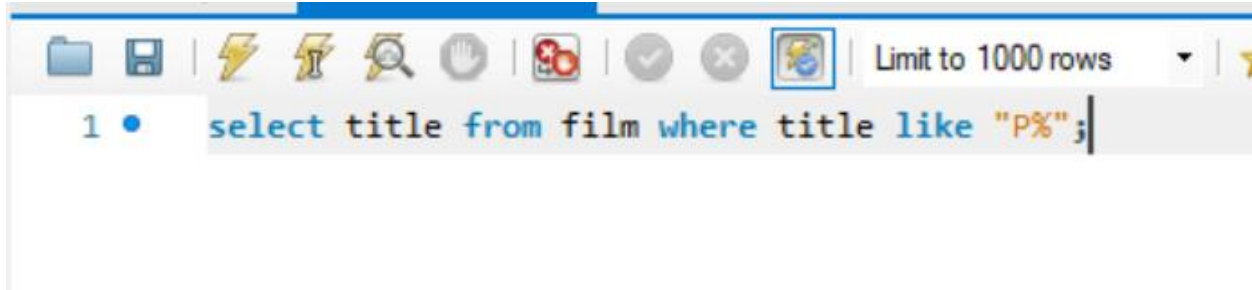
OUTPUT:

	actor_id
▶	58
✱	NULL



2. Write a query to retrieve names of movies starting with P.

QUERY:



OUTPUT:

Result Grid				Filter Rows:
	title			
▶	PACIFIC AMISTAD			
	PACKER MADIGAN			
	PAJAMA JAWBREAKER			
	PANIC CLUB			
	PANKY SUBMARINE			
	PANTHER REDS			



3. Write a query to retrieve movies which were released in year 2006?

QUERY:



OUTPUT:

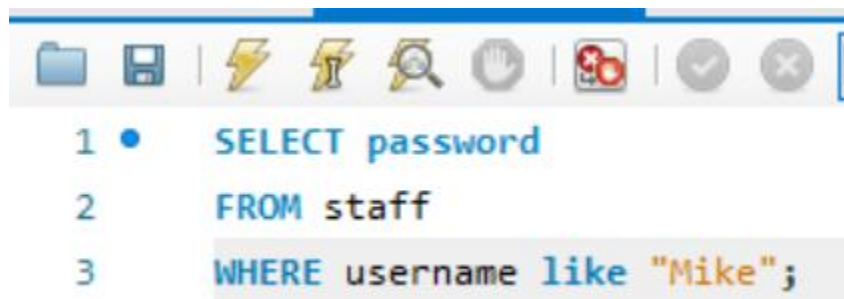
A screenshot of a database result grid. The grid has a header row with the column name "title". Below the header, there are several rows of movie titles. The first row is "ACADEMY DINOSAUR". The second row is "ACE GOLDFINGER". The third row is "ADAPTATION HOLES". The fourth row is "AFFAIR PREJUDICE". The fifth row is "AFRICAN EGG". The sixth row is "AGENT TRUMAN". The seventh row is "AFRICAN EGG". The eighth row is "AGENT TRUMAN". The ninth row is "AFRICAN EGG". The tenth row is "AGENT TRUMAN".

	title
▶	ACADEMY DINOSAUR
	ACE GOLDFINGER
	ADAPTATION HOLES
	AFFAIR PREJUDICE
	AFRICAN EGG
	AGENT TRUMAN
	AFRICAN EGG
	AGENT TRUMAN
	AFRICAN EGG
	AGENT TRUMAN



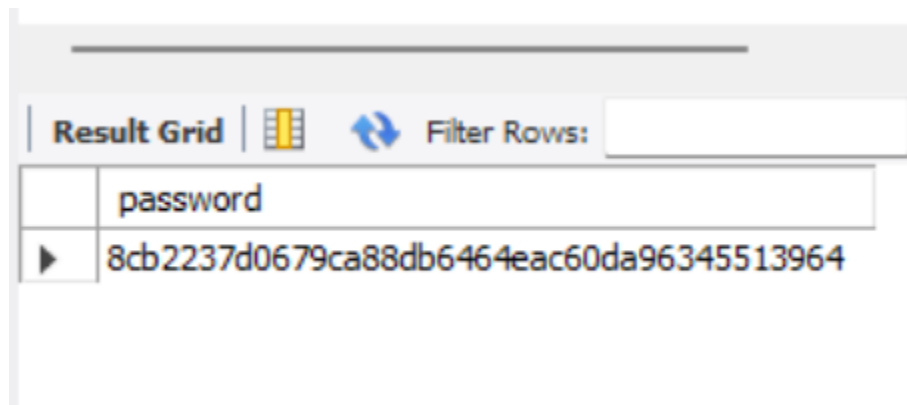
4. What password did the DBA assign to the user „MIKE“?

QUERY:



```
1 • SELECT password
2 FROM staff
3 WHERE username like "Mike";
```

OUTPUT:



	password
▶	8cb2237d0679ca88db6464eac60da96345513964



5. Write a query to retrieve data of all actors whose first names are not ending on T.

QUERY:

```
1 • SELECT
2 *FROM actor
3 WHERE first_name not like "%t";
```

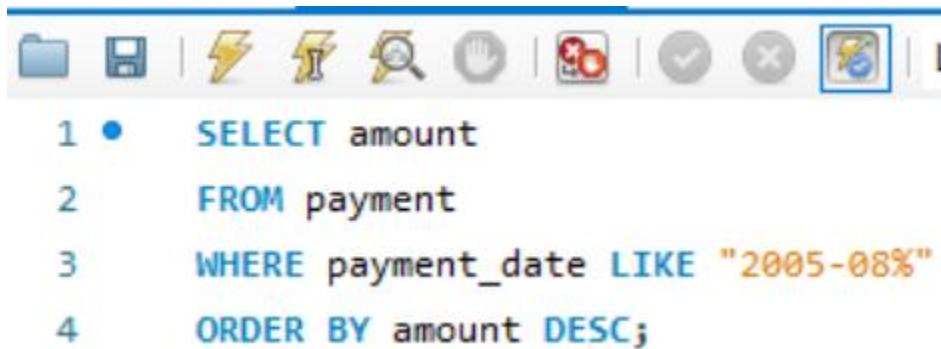
OUTPUT:

Result Grid				
Filter Rows:				
Edit:				
	actor_id	first_name	last_name	last_update
▶	1	PENELOPE	GUINNESS	2006-02-15 04:34:33
	2	NICK	WAHLBERG	2006-02-15 04:34:33
	3	ED	CHASE	2006-02-15 04:34:33
	4	JENNIFER	DAVIS	2006-02-15 04:34:33
	5	JOHNNY	LOLLOBRIGIDA	2006-02-15 04:34:33
	6	BETTE	NICHOLSON	2006-02-15 04:34:33
	7	COLLEEN	MOORE	2006-02-15 04:34:33



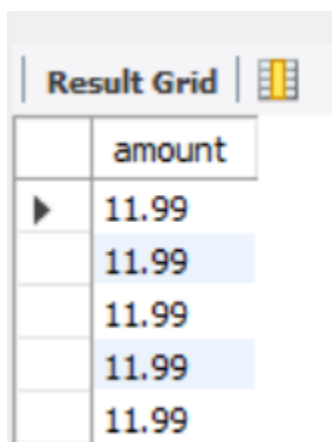
6. Find and display the Income (payments) generated during August 2005.
Sort them in the descending order.

QUERY:



```
1 • SELECT amount
2 FROM payment
3 WHERE payment_date LIKE "2005-08%"
4 ORDER BY amount DESC;
```

OUTPUT:

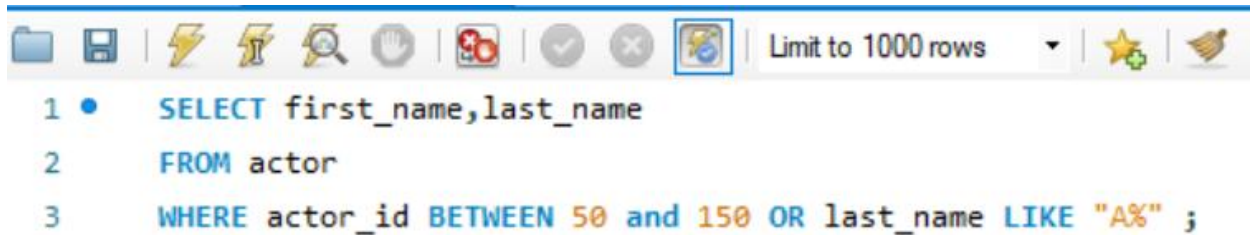


	amount
▶	11.99
	11.99
	11.99
	11.99
	11.99



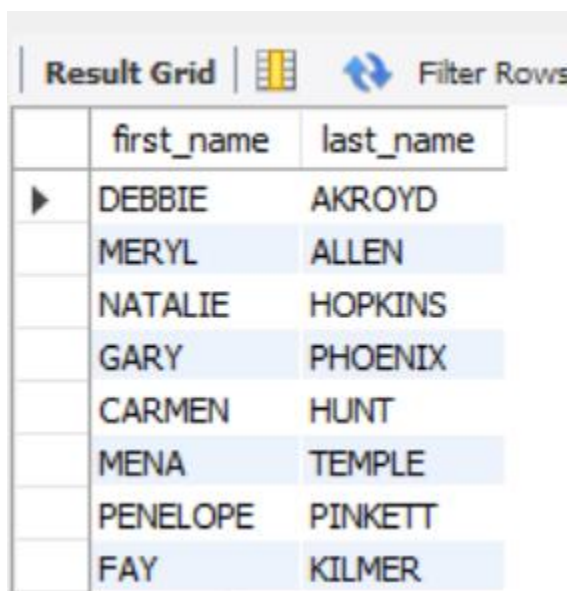
7. Select the names of actors whose IDs are between 50 and 150, or those whose last name starts with A.

QUERY:



```
1 • SELECT first_name, last_name
2 FROM actor
3 WHERE actor_id BETWEEN 50 and 150 OR last_name LIKE "A%" ;
```

OUTPUT:

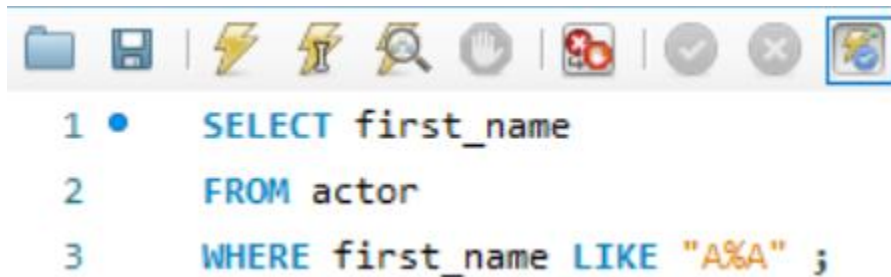


	first_name	last_name
▶	DEBBIE	AKROYD
	MERYL	ALLEN
	NATALIE	HOPKINS
	GARY	PHOENIX
	CARMEN	HUNT
	MENA	TEMPLE
	PENELOPE	PINKETT
	FAY	KILMER



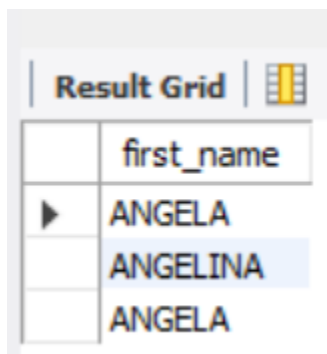
8. Display the actor first name which start with A and end on letter A.

QUERY:



```
1 • SELECT first_name
2   FROM actor
3   WHERE first_name LIKE "A%A" ;
```

OUTPUT:



	first_name
▶	ANGELA
	ANGELINA
	ANGELA



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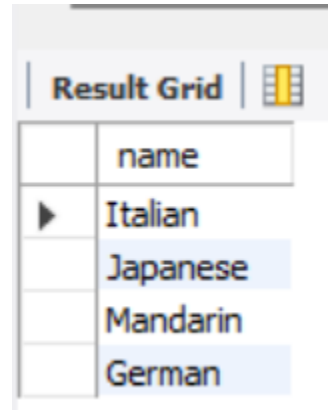
9. Write an SQL query to display list of languages which neither is English nor French.

QUERY:



```
1 • SELECT name
2 FROM language
3 WHERE name != "English" AND name != "French";
```

OUTPUT:



	name
▶	Italian
	Japanese
	Mandarin
	German

Deliverables:

Submit a word document including the SQL queries to answer above-mentioned information needs as well as snapshot of their outcome when executed over MySQL using the Workbench.



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